

A Hybrid Vision-RTK Navigation Framework for Autonomous Spatio-Temporal Data Collection in Dense Vineyards

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Abstract—Precision viticulture is dependent on robust spatio-temporal data, but its acquisition is hampered by unreliable autonomous navigation in challenging vineyard environments. This paper presents a robotic system for spatio-temporal data collection, built upon a robust and low-cost navigation framework. The framework integrates a stereo depth camera for precise, vision-based in-row guidance with an RTK-GNSS system for global waypoint management and row transitions between lanes. The algorithms were tested in a high fidelity simulation environment and the system’s efficacy is demonstrated through a targeted data collection by deploying the same on a ground rover with custom-designed sensor rig with side-mounted cameras to systematically capture crop imagery in a dense vineyard validating autonomous navigation and the system’s functionality as an end-to-end data acquisition tool. This work provides a validated, cost-effective platform for automated data acquisition, which can be extended to other precision agriculture tasks.

Index Terms—Viticulture, precision agriculture, vineyard, autonomous navigation, agricultural robot, robotics.

I. INTRODUCTION

Sustainable agricultural productivity is very important to address the global challenges of a rising population, climate change, and a shrinking labor force [1]. In this context, Precision Agriculture (PA) has emerged as a critical paradigm, which leverages technology to create Decision Support Systems (DSS) that optimize farming operations. Autonomous robots are at the forefront of this technological wave, which promises to automate labor-intensive tasks and enable data-driven management [2]. These systems are particularly impactful in specialized domains such as Precision Viticulture (PV), where technology is used to monitor and improve grape quality and yield [3]. The successful deployment of robots for such tasks fundamentally relies on the ability to navigate robustly and map farmland accurately, a key area of ongoing research [4], [5], [6].

Viticulture demands significant time and manual effort, relying heavily on specifically skilled labor throughout its growing cycle. However, with the increasing shortage of available workers, there is a growing push toward automation

to maintain productivity and efficiency in vineyard operations. India is a major table grape producer [7] and relies heavily on manual labor, which can lead to inefficiencies and inconsistent practices. Therefore, developing autonomous systems for vineyards offers a direct path to increasing efficiency, improving crop consistency, and reducing operational costs.

The vineyard environment is semi-structured and dynamic. Dense fruit canopies frequently obstruct Global Navigation Satellite System (GNSS) signals, making standard GPS unreliable for precise in-row maneuvers. The challenges of GNSS signal degradation and multipath errors in such settings are well-documented by Rovira et. al. [8]. While high-precision systems like Real-Time Kinematic (RTK) GNSS offer centimeter-level accuracy in open-sky conditions, but their performance degrades significantly under dense canopies. Also, GNSS-based navigation relies on an Inertial Measurement Unit (IMU) for heading, and as IMUs inherently accu-

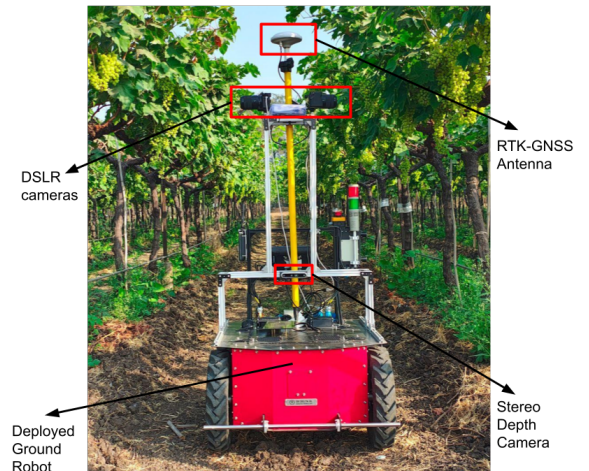


Fig. 1. Vineyard navigation robot

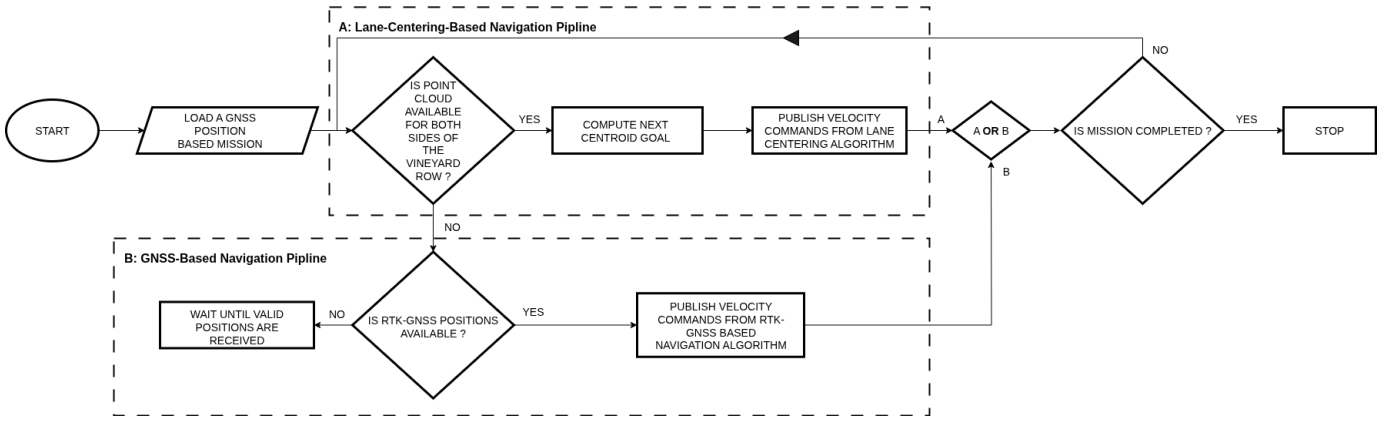


Fig. 2. Navigation flowchart

multitude drift error over time, a problem exacerbated by uneven farm terrain. In addition, India's vineyards are more dense compared to international standards to obtain higher yields per acre, making the solution more complex.

To overcome these challenges, the state-of-the-art has largely focused on augmenting GNSS with environment-perception sensors, primarily LiDAR (mostly 3D LiDAR). The work by Mengoli et al. [9] presents a versatile robotic platform that uses RTK-GNSS for "open-field" travel and switches to a LiDAR-based algorithm for "in-row" guidance. Yet, these solutions present a clear trade-off. High-end 3D LiDAR sensors, while providing rich point cloud data [12], are very expensive. This has led some researchers to explore more cost-effective perception methods, like 2D LiDAR and 2D camera as a viable alternative for 3D perception [10], [11]. A research gap exists for a solution that leverages a low-cost perception sensor in conjunction with GNSS to achieve high reliability without the associated cost.

This paper presents a low-cost, robust hybrid navigation framework for an autonomous agricultural robot, designed and validated for spatio-temporal data collection in a dense vineyard to overcome the challenges mentioned above. Instead of sensor fusion, our system employs a resilient, dual-algorithm approach that integrates the capabilities of a stereo depth camera and an RTK-GNSS system. The two systems operate concurrently: a vision-based, lane-centering algorithm uses dense 3D point cloud data from the stereo camera for precise in-row guidance. In parallel, the RTK-GNSS provides high-accuracy global positioning to manage the overall mission, execute row-end turns, and also take over navigation control if the visual data becomes unavailable. The proposed algorithms were developed and experimentally verified using a robotic platform, as depicted in Fig. 1. This four-wheel drive system features dimensions of 950 mm (L) $\times$ 800 mm (W) $\times$ 480 mm (H). The sensory setup includes an Intel RealSense D455 RGB-D camera mounted at the front for depth and color imaging, a u-blox ZED-F9P GNSS receiver enabling high-precision RTK-based navigation, and two Sony Alpha 6700 cameras utilized for data collection.

The main contributions of this paper are:

- 1) A low-cost hybrid navigation framework combining a stereo depth camera and RTK-GNSS, validated in a high-fidelity ROS/Gazebo simulation before field deployment.
- 2) A robust lane-centering algorithm using filtered point cloud data for precise in-row positioning.
- 3) An autonomous mode-switching strategy between vision-based and GNSS-based navigation for continuous operation.
- 4) Full field validation in a dense vineyard as an end-to-end spatio-temporal data collection tool.

The system switches between a vision-based **Local Planner** for in-row guidance (Section II) and an RTK-GNSS-based **Global Planner** for waypoint navigation and row transitions (Section III), as illustrated in Figure 2. After acquiring a valid RTK fix, the robot enters a continuous control loop: the Local Planner is active when stable point cloud data is available; otherwise the Global Planner navigates via pre-loaded waypoints until visual data is restored.

The remainder of this paper is organized as follows. Section II details the vision-based lane centering algorithm, covering its development in a high-fidelity ROS [13] and Gazebo [14] simulation and an analysis of its tracking performance. Section III presents the complementary RTK-GNSS based navigation system, which serves as the global planner and robust fallback. The primary application of this framework for systematic data acquisition which is described in Section IV. Finally, Section V concludes the paper by summarizing our key contributions and suggesting avenues for future research.

II. VISION BASED LANE CENTERING

To validate our navigation framework, we adopted a two-stage approach: initial development and testing in a high-fidelity simulation environment, followed by deployment and performance analysis on the physical robot in a real-world vineyard. This methodology allowed for rapid, safe, and cost-effective iteration of the core algorithms before field trials.

A. Simulation Environment

The simulation was built using ROS 1 Noetic [13] and Gazebo [14], as shown in Figure 3. The simulated robot matches the physical platform’s dimensions, kinematics, and sensor suite including an RGB-D camera, 2D LiDAR, IMU, and wheel odometry ensuring direct sim-to-real transfer of perception algorithms. The environment features undulating terrain and equidistant vineyard rows to validate lane-following and row-switching logic.

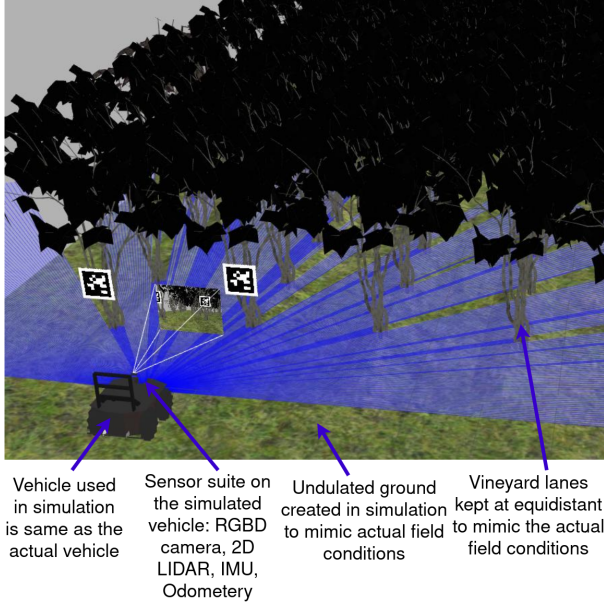


Fig. 3. The custom Gazebo simulation environment. It features a robot model with the same dimensions and sensor suite as the physical platform, operating in a world with procedurally generated undulating terrain and structured crop rows to mimic real-world conditions.

The repository containing the environment and its implementation is shared on Github [18]

B. Lane Centering Algorithm

Maintaining the robot’s alignment at the center of the vineyard row is essential for accurate data collection and safe operation. This is achieved through the lane centering algorithm, which processes 3D point cloud data to compute a stable navigation waypoint. The algorithm’s performance, both in simulation and during field trials, is visualized in Figure 4. The following subsections outline the algorithm’s pipeline in detail.

1) *Point Cloud Preprocessing*: The computation begins with the raw point cloud $P = \{p_i = (x_i, y_i, z_i) \in \mathbb{R}^3\}_{i=1}^n$ from the stereo depth camera. As processing the full cloud is computationally intensive, a two-step preprocessing pipeline is performed:

- 1) **Downsampling**: The point cloud is uniformly downsampled by a configurable factor to reduce the number of points for subsequent processing.
- 2) **Spatial Filtering**: A 3D cuboid filter, or Region of Interest (ROI), is applied to isolate points corresponding

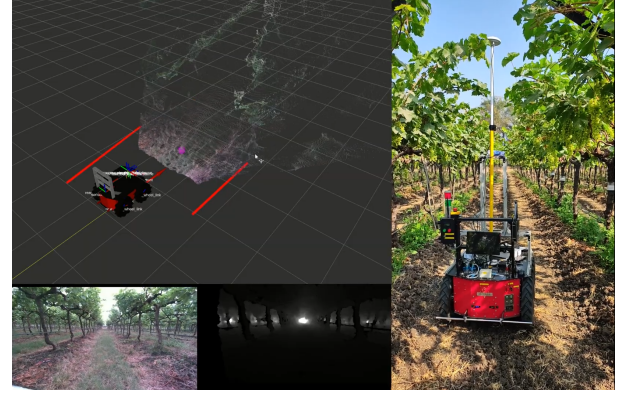


Fig. 4. Composite view of the system in operation. (Right) The physical robot in a dense vineyard. (Top-Left) The corresponding RViz visualization showing the robot model, the raw point cloud, and the algorithm’s output. The robot’s forward-facing RGB camera view and the depth camera’s view are shown bottom left sequentially.

to the vine trunks and lower canopy. This removes the ground plane, the upper canopy, and irrelevant objects outside the immediate path. The filtered point cloud, P_{filtered} , is defined as:

$$P_{\text{filtered}} = \{p_i \in P \mid x \in [w_{\min}, w_{\max}], y \in [h_{\min}, h_{\max}], z \in [d_{\min}, d_{\max}]\} \quad (1)$$

where $[w_{\min}, w_{\max}]$ defines the lateral width, $[h_{\min}, h_{\max}]$ defines the vertical height, and $[d_{\min}, d_{\max}]$ defines the depth range relative to the camera frame. This ROI is visualized as the semi-transparent green cuboid in Figure 5. These parameters are configurable and can be changed in the corresponding ROS nodes.

2) *Lane Boundary Detection and Clustering*: The filtered points are then processed to identify the lane boundaries:

- 1) **Lateral Separation**: P_{filtered} is split into two subsets based on their position relative to the camera’s central axis: P_{left} and P_{right} .
- 2) **Clustering**: The **DBSCAN** [15] (Density-Based Spatial Clustering of Applications with Noise) algorithm is applied independently to P_{left} and P_{right} . DBSCAN was chosen for its ability to find arbitrarily shaped clusters and its robustness to noise, making it well-suited for unstructured natural environments.

$$C_k = \text{DBSCAN}(P_k, \epsilon, m_{\min}), \quad k \in \{\text{left}, \text{right}\} \quad (2)$$

In this context, ϵ defines the maximum allowable distance between two samples for them to be considered neighbors. Meanwhile, $m_{\min}$ represents the minimum number of samples in a neighborhood for a point to qualify as a core point.

3) *Lane State Estimation and Waypoint Projection*: The clusters generated by DBSCAN are used to compute the final navigation goal. The process involves centroid extraction,

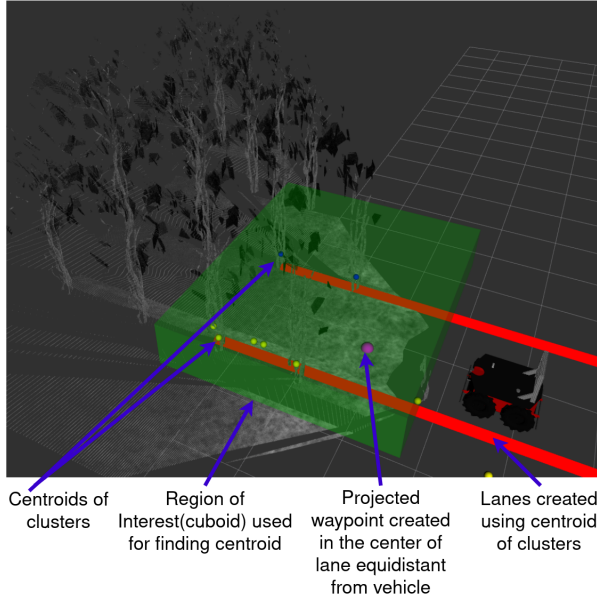


Fig. 5. RViz visualization of the lane centering algorithm's output. The yellow and blue spheres are the centroids of detected clusters. The green cuboid represents the Region of Interest (ROI) for point cloud filtering. The red lines represent the final computed lane boundaries, and the purple sphere is the projected waypoint sent to the robot's controller.

coordinate transformation, and center line definition, culminating in the projection of a target waypoint, as detailed in Algorithm 1. This projection method, a key feature of our approach, finds the point on the lane's center line closest to the robot and then projects a new waypoint a fixed look-ahead distance, d , from that point. The final projected point, $\mathbf{p}_{\text{waypoint}}$, is published as the target waypoint for the robot's controller.

Algorithm 1 Trajectory Point Projection

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1: function PROJECTWAYPOINT( $\mathbf{p}_{\text{start}}$ ,  $\mathbf{p}_{\text{end}}$ ,  $\mathbf{p}_{\text{robot}}$ ,  $d$ )
2:    $\mathbf{v} \leftarrow \mathbf{p}_{\text{end}} - \mathbf{p}_{\text{start}}$   $\triangleright$  Lane direction vector
3:    $\mathbf{w} \leftarrow \mathbf{p}_{\text{robot}} - \mathbf{p}_{\text{start}}$ 
4:    $t \leftarrow (\mathbf{w} \cdot \mathbf{v}) / (\mathbf{v} \cdot \mathbf{v})$   $\triangleright$  Projection factor
5:    $\mathbf{p}_{\text{closest}} \leftarrow \mathbf{p}_{\text{start}} + t \cdot \mathbf{v}$   $\triangleright$  Closest point on line to robot
6:    $\hat{\mathbf{v}} \leftarrow \mathbf{v} / \|\mathbf{v}\|$   $\triangleright$  Unit direction vector
7:    $\mathbf{p}_{\text{waypoint}} \leftarrow \mathbf{p}_{\text{closest}} + d \cdot \hat{\mathbf{v}}$   $\triangleright$  Project look-ahead point
8:   return  $\mathbf{p}_{\text{waypoint}}$ 
9: end function

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4) *Waypoint Following Controller:* For robot actuation, we implemented a Pure Pursuit-style look-ahead controller. This controller subscribes to the projected waypoint and the robot's current odometry. In each control cycle, it calculates the heading error between the robot's current orientation and the bearing to the waypoint. It then commands a constant linear velocity and an angular velocity that is directly proportional to this heading error.

C. Performance Analysis

Figure 6 shows the desired path alongside the robot's actual trajectory over a 20-meter run. The desired path (red

dashed) exhibits high-frequency oscillations characteristic of noisy real-time perception, while the actual path (solid blue) is visibly smoother — the controller and physical dynamics act as a low-pass filter. The lateral standard deviation of the actual trajectory was **0.12m**, confirming stable in-row tracking.

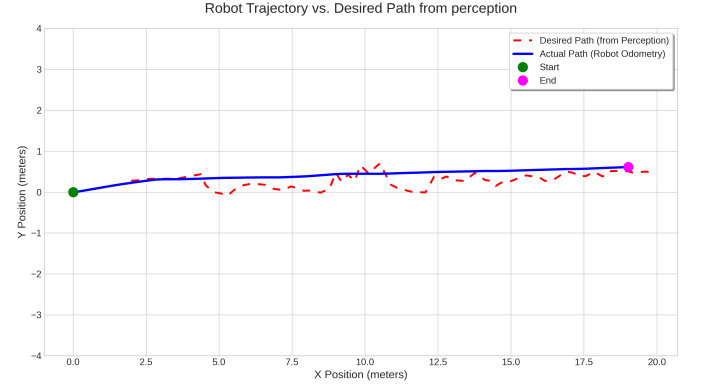


Fig. 6. Desired path from the perception algorithm vs. the robot's actual trajectory from odometry. The controller effectively smooths the noisy perception output, resulting in a stable path.

The tracking MAE and RMSE of 1.824m and 1.836m respectively reflect the look-ahead controller's intentional 2.0m waypoint projection ahead of the robot, not lateral lane deviation confirming the controller operates as designed. The algorithm was subsequently deployed in the field (Figure 4) with comparable results.

III. RTK-GNSS BASED AUTONOMOUS NAVIGATION

The Real-Time Kinematic Global Navigation Satellite System (RTK-GNSS) plays a crucial role in autonomous navigation in the agriculture field. It is used to mark the waypoints for navigation, assist in switching vineyard lanes during autonomous navigation, and assist the lane-centering algorithm when no point clouds are available on either side of the rows. The aim is to develop a reliable solution for precision agriculture. Initially, for GNSS-based navigation, we evaluated the autonomous navigation with only GNSS and IMU. The results showed that the deviation between the desired and the actual was a bit more than 7.0 meters, refer Figure 7, which is not desirable, as in the case of the actual field, the distance between two adjacent rows is less than 3 meters.

To overcome these challenge, a Continuously Operating Reference Stations (CORS) service offered by the Survey of India [16] was evaluated. In this method, an RTK-GNSS sensor, when connected to a server, receives RTCM correction data, which helped to achieve centimeter-level accuracy within a few minutes. For this to work, a stable internet connection and a GNSS receiver that supports RTK (Real-time Kinematic) are required. Using this configuration, autonomous navigation was performed to navigate through 6 vineyard lanes in a fully dense vineyard canopy. The results showed that with the RTK-GNSS, we were able to achieve an accuracy of less than 1.0 meter shown in Figure 8. This error also includes the drift error, which gets accumulated in the Inertial Measurement

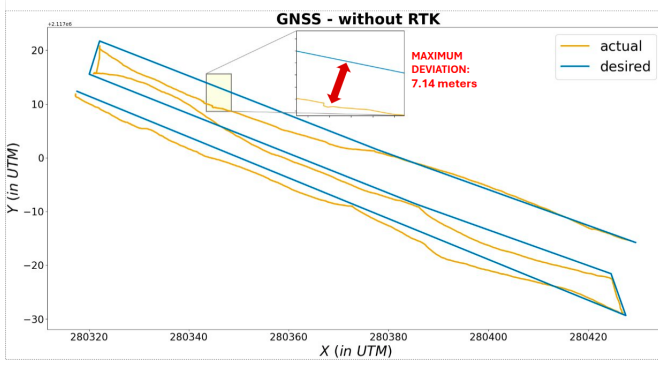


Fig. 7. Non RTK-GNSS based navigation

Unit (IMU). Another experiment was conducted in which the algorithm computes interpolated points between the start and the goal position. The idea for was to correct the motion at every set distance, but it turns out to add even more drift in the IMU.

The RTK-GNSS-based navigation algorithm assist the robot in navigating through all the marked waypoints. The current location of the robot is known with the help of a GNSS sensor. Euclidean distance and heading are computed between the current position and the desired waypoints. The errors obtained from this is fed to correct the linear and angular error of a robot. Please refer pseudo code in Algorithm 2 for more details. In the proposed algorithm, geographic coordinates-typically expressed in latitude and longitude are transformed into a universal transverse mercator (UTM) coordinate system. As latitude and longitude define positions on a spherical surface, they are not ideal for the planar computations required in ground robot navigation. Converting these coordinates into UTM, which provides a two-dimensional Cartesian representation (Easting and Northing), allows for more straightforward spatial processing, enhancing the accuracy and efficiency of the navigation system [17].

The integrated RTK-GNSS-based navigation and lane-centering algorithm operates concurrently, each compensating for the limitations of the other to enhance overall system reliability and accuracy. Additionally, to switch lanes at the end of each row, only the RTK-GNSS-based navigation remains active. The flow of the complete autonomous navigation algorithm is shown in Figure 2.

The autonomous navigation system, implemented using Algorithm 2, was quantitatively evaluated in both non-RTK and RTK-GNSS modes. Without RTK, the system showed a maximum deviation of 7.14 meters, with a mean of 1.432 meters and a standard deviation of 0.929 meters (Figure 7). With RTK-GNSS, accuracy improved significantly, reducing the maximum deviation to 0.88 meters, with a mean of 0.297 meters and a standard deviation of 0.174 meters (Figure 8). A key point to note is that lane switching and end-of-row turns are handled exclusively by the RTK-GNSS system, underscoring its role in precise navigation.

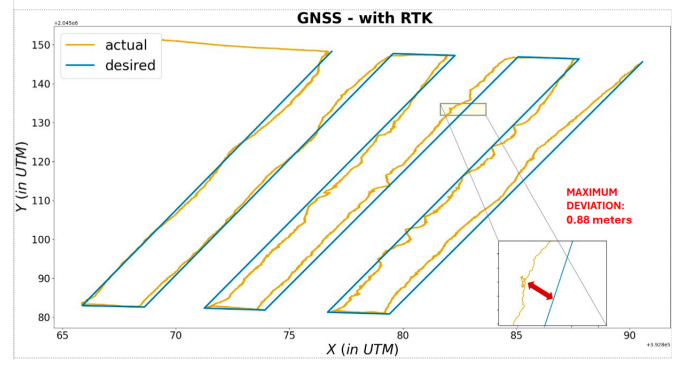


Fig. 8. RTK-GNSS based navigation

A. Complete RTK-GNSS-based navigation pseudo code

Algorithm 2 RTK-GNSS Based Autonomous Navigation

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1: Load goal positions  $\{(E_i, N_i)\}$  from CSV
2: for each goal  $(E_d, N_d)$  do
3:   Initialize controller and PID variables
4:   while robot active and goal not reached do
5:     if GNSS data is valid then
6:       Convert GNSS coordinates to UTM:  $(E, N)$ 
7:       Compute distance and heading errors:  $e_v, e_\omega$ 
8:       if RTK Fix available then
9:         if  $e_\omega > \text{threshold}$  then
10:          Rotate to correct heading
11:        else
12:          Move linearly toward the goal
13:        end if
14:        if goal and heading error  $<$  thresholds then
15:          break
16:        end if
17:      else
18:        Wait for valid RTK fix
19:      end if
20:    else
21:      Wait until valid GNSS data is received
22:    end if
23:    Send velocities, update previous errors and time
24:  end while
25: end for
26: Stop robot:  $v_{\text{lin}} = 0, \omega = 0$ 

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IV. DATA ACQUISITION

Two cameras are on the rig of the robot on lateral side(either side) of the robot as shown in figure 1. Cameras are triggered at fixed travel intervals using wheel odometry, ensuring consistent spatial sampling for downstream tasks such as image stitching and structure-from-motion (SfM). The trigger distance d is derived from the camera FOV and perpendicular distance to the canopy d_{tc} , with desired overlap X :

$$d = 2 \cdot d_{tc} \cdot \tan\left(\frac{\text{FOV}}{2}\right) \cdot \left(1 - \frac{X}{100}\right) \quad (3)$$

The parameters are explained in Figure 9

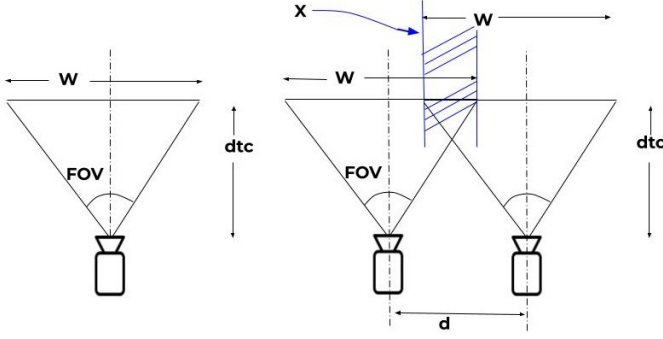


Fig. 9. Geometric model for image overlap control. W : coverage width at canopy; d_{tc} : camera-to-canopy distance; **FOV**: horizontal field of view; d : inter-capture travel distance; X : desired overlap.

In deployment, d_{tc} was supplied directly by the lane-centering algorithm and FOV was a known camera constant. With $X = 30\%$, the computed d was used to trigger the side-mounted Sony Alpha 6700 cameras via odometry, producing a geometrically consistent dataset along each vineyard row. Along with images we also generate a meta data file containing the GPS tag, timestamp and nomenclature of image along with overlap details to be used during post-processing.

V. CONCLUSION

This paper presented a low-cost hybrid navigation framework for autonomous data collection in dense Indian vineyards, where tight inter-row spacing and uneven terrain make navigation particularly challenging. The dual-mode system — combining stereo vision-based lane centering (lateral deviation: 0.12m) with CORS-RTK-GNSS fallback (max deviation: 0.88m) — was validated in simulation and field-deployed for systematic crop imagery acquisition using side-mounted cameras. The framework's cost-effectiveness, redundancy, and low computational overhead make it directly extensible to other precision agriculture tasks including spraying, seeding, and harvesting.

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REFERENCES

- [1] FAO. 2017. "The future of food and agriculture – Trends and challenges" Food and Agriculture Organization of the United Nations, Rome, 163 p.
- [2] Shamshiri R R, Weltzien C, Hameed I A, Yule I J, Grift T E, Balasundram S K, et al. Research and development in agricultural robotics: A perspective of digital farming. *Int J Agric and Biol Eng*, 2018; 11(4): 1–14.

- [3] Ammoniaci, M., Kartsiotis, S.-P., Perria, R., and Storch, P. (2021). State of the Art of Monitoring Technologies and Data Processing for Precision Viticulture. *Agriculture*, 11(3), 201. <https://doi.org/10.3390/agriculture11030201>
- [4] Diego Tiozzo Fasiolo, Lorenzo Scalera, Eleonora Maset, Alessandro Gasparetto, Towards autonomous mapping in agriculture: A review of supportive technologies for ground robotics, *Robotics and Autonomous Systems*, Volume 169, 2023, 104514, ISSN 0921-8890, <https://doi.org/10.1016/j.robot.2023.104514>.
- [5] Ali Roshanianfard, Noboru Noguchi, Hiroshi Okamoto, Kazunobu Ishii, A review of autonomous agricultural vehicles (The experience of Hokkaido University), *Journal of Terramechanics*, Volume 91, 2020, Pages 155-183, ISSN 0022-4898, <https://doi.org/10.1016/j.jterra.2020.06.006>.
- [6] Binbin Xie, Yucheng Jin, Muhammad Faheem, Wenjie Gao, Jizhan Liu, Houkang Jiang, Lianjiang Cai, Yuanxiang Li, Research progress of autonomous navigation technology for multi-agricultural scenes, *Computers and Electronics in Agriculture*, Volume 211, 2023, 107963, ISSN 0168-1699, <https://doi.org/10.1016/j.compag.2023.107963>.
- [7] U.S. Department of Agriculture, Foreign Agricultural Service, *Production, Commodity: Grapes*. [Online]. Available: <https://www.fas.usda.gov/data/production/commodity/0575100> [Accessed: Jul. 28, 2025].
- [8] Rovira-Más, F. Sensor Architecture and Task Classification for Agricultural Vehicles and Environments. *Sensors* 2010, 10, 11226-11247. <https://doi.org/10.3390/s101211226>
- [9] Mengoli, Dario & Tazzari, Roberto & Marconi, Lorenzo. (2020). Autonomous Robotic Platform for Precision Orchard Management: Architecture and Software Perspective. 303-308. 10.1109/MetroAgri-For50201.2020.9277555.
- [10] S. Thamaraiselvan, M. I. Vivin, R. K. R. Ray and B. D, "LIDAR-based Navigation Rover for Fields with Smart Pest Sprayer using Machine Vision," 2023 9th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 2023, pp. 1889-1894, doi: 10.1109/ICACCS57279.2023.10113107.
- [11] Rajitha de Silva, Grzegorz Cielniak, Junfeng Gao, Vision based crop row navigation under varying field conditions in arable fields, *Computers and Electronics in Agriculture*, Volume 217, 2024, 108581, ISSN 0168-1699, <https://doi.org/10.1016/j.compag.2023.108581>.
- [12] Tuan Le, Jon Glenn Omholt Gjevestad, Pål Johan From, Online 3D Mapping and Localization System for Agricultural Robots, *IFAC-PapersOnLine*, Volume 52, Issue 30, 2019, Pages 167-172, ISSN 2405-8963, <https://doi.org/10.1016/j.ifacol.2019.12.516>.
- [13] Quigley, Morgan & Conley, Ken & Gerkey, Brian & Faust, Josh & Foote, Tully & Leibs, Jeremy & Wheeler, Rob & Ng, Andrew. (2009). ROS: an open-source Robot Operating System. ICRA Workshop on Open Source Software. 3.
- [14] N. Koenig and A. Howard. Design and use paradigms for gazebo, an open-source multi-robot simulator. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 2149-2154, Sendai, Japan, Sep 2004.
- [15] Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu. 1996. A density-based algorithm for discovering clusters in large spatial databases with noise. In *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD'96)*. AAAI Press, 226–231.
- [16] Survey of India, *Continuously Operating Reference Stations (CORS) – Introduction*. [Online]. Available: <https://cors.surveyofindia.gov.in/introduction> [Accessed: Jul. 29, 2025]. In: IEEE International Conference on Robotics and Automation (ICAR 2023).
- [17] Yilong Li, Qingchun Feng, Chao Ji, Jiahui Sun and Yu Sun: GNSS and LiDAR Integrated Navigation Methods in Orchards with Intermittent GNSS Dropout. In: *Appl. Sci.* 2024,.
- [18] https://github.com/Shreyash007/Vineyard_navigation_robot